00_vehicle_input
: 00_vehicle_input.kicad_sch

00b_reverse_ov
: 00b_reverse_ov.kicad_sch

00c_auto_5v
: 00c_auto_5v.kicad_sch

01_usb_c
: 01_usb_c.kicad_sch

01a_usb_cc_supply
: 01a_usb_cc_supply.kicad_sch

01b_usb_limit
: 01b_usb_limit.kicad_sch

02_system_power
: 02_system_power.kicad_sch

03_reset
: 03_reset.kicad_sch

03b_phy_power
: 03b_phy_power.kicad_sch

03c_auto_qualification
: 03c_auto_qualification.kicad_sch

19_gps_power
: 19_gps_power.kicad_sch

04_stm32
: 04_stm32.kicad_sch

05_debug_clock_uart
: 05_debug_clock_uart.kicad_sch

06_esp32
: 06_esp32.kicad_sch

07_esp_service
: 07_esp_service.kicad_sch

08_can_fd
: 08_can_fd.kicad_sch

09_can_ft
: 09_can_ft.kicad_sch

10_tx_gate
: 10_tx_gate.kicad_sch

11_phy_modes
: 11_phy_modes.kicad_sch

12_watchdog_latch
: 12_watchdog_latch.kicad_sch

12b_latch_clear
: 12b_latch_clear.kicad_sch

13_arm_switch
: 13_arm_switch.kicad_sch

14_can_connectors
: 14_can_connectors.kicad_sch

15_gnss_ins
: 15_gnss_ins.kicad_sch

16_barometer
: 16_barometer.kicad_sch

17_gps_ports
: 17_gps_ports.kicad_sch

18_gps_protection
: 18_gps_protection.kicad_sch

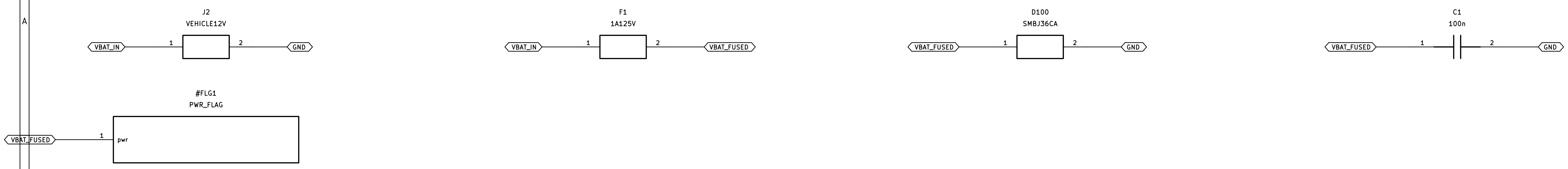
20_test_points
: 20_test_points.kicad_sch

SCHEMATIC REVIEW ONLY — PCB/HIL release gates remain open
digitie/canview
Sheet: /
File: communicator.kicad_sch

Title: CANView / communicator

Size: A3 | Date: 2026-09-05 | **Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6 | Id: 1/29

12V fused branch only. Common-source LM74800, bidirectional input TVS. No claim of validated unsuppressed load-dump survival.



SCHEMATIC REVIEW ONLY – PCB/HIL release gates remain open

digitie/canview

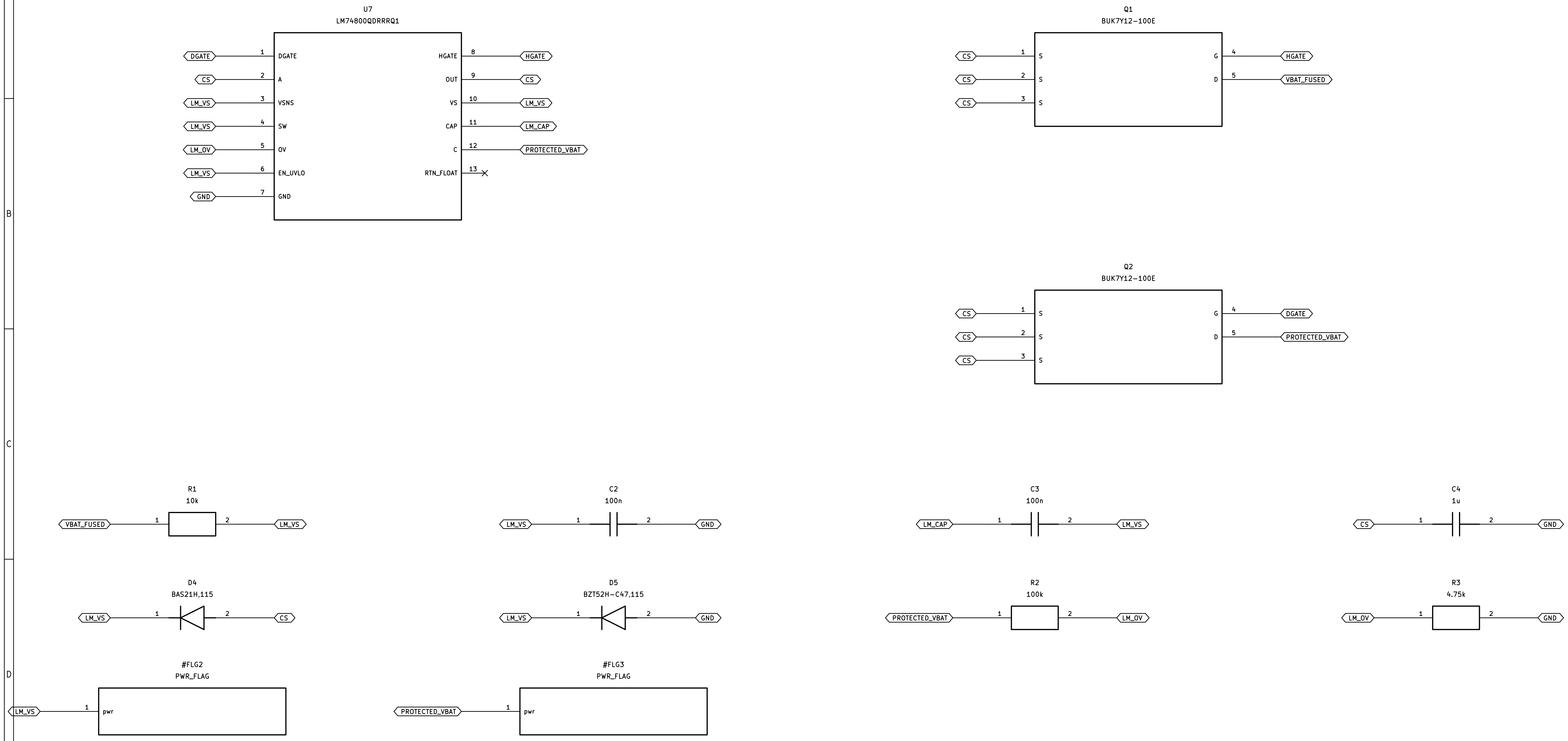
Sheet: /00_vehicle_input/

File: 00_vehicle_input.kicad_sch

Title: CANView / 00_vehicle_input

Size: A3	Date: 2026-09-05	**Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 2/29

LM74800 common-source: HGATE drives INPUT FET; DGATE drives OUTPUT FET; A/OUT both=CS. Exposed pad FLOATS.



SCHEMATIC REVIEW ONLY - PCB/HIL release gates remain open

digitie/canview

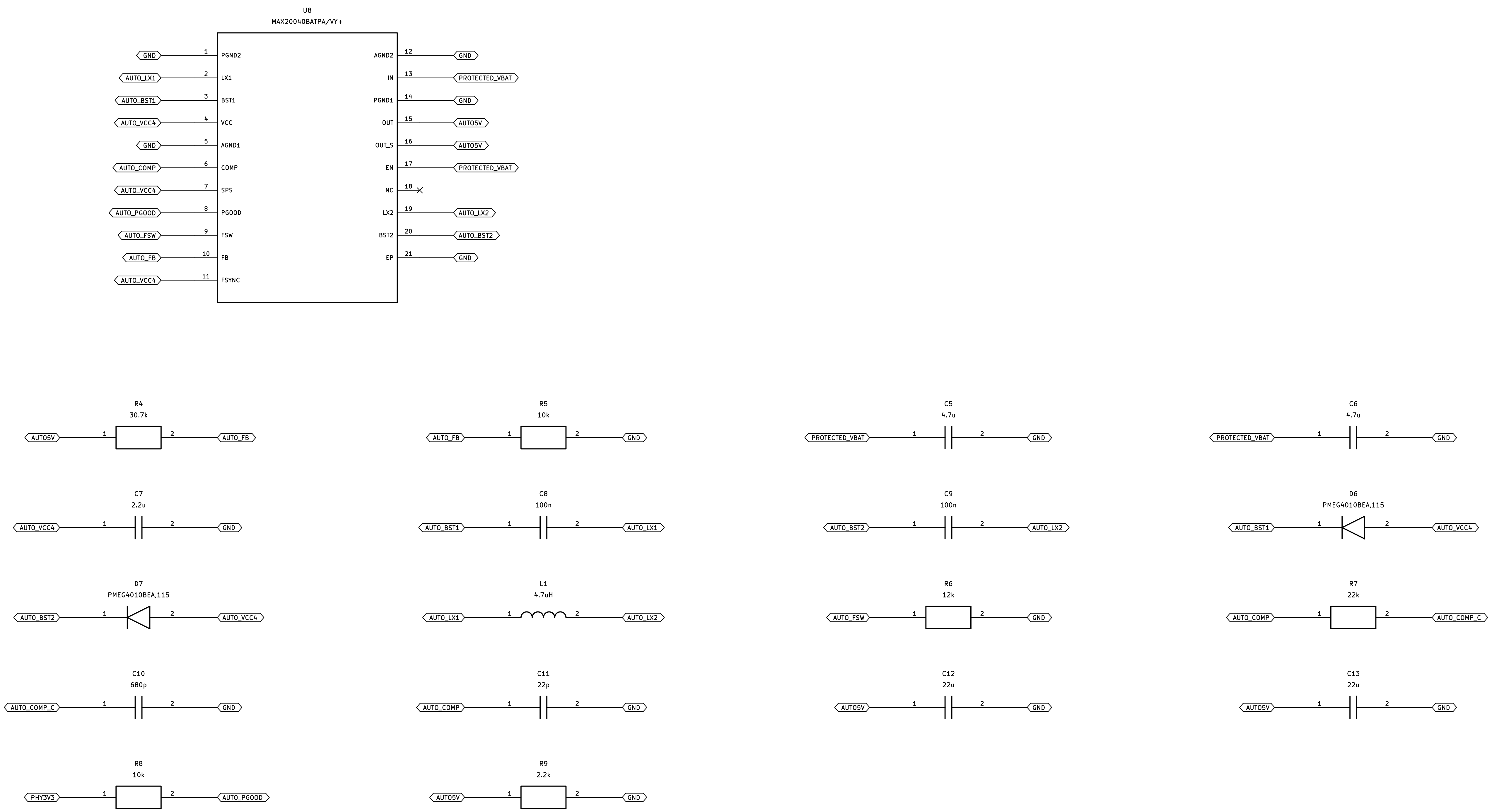
Sheet: /00b_reverse_ov/

File: 00b_reverse_ov.kicad_sch

Title: CANView / 00b_reverse_ov

Size: A3	Date: 2026-09-05	**Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 3/29

MAX20040B adjustable5.0875V at2.2MHz, inside PHY5V range. OUT_S Kelvin.4.5V startup; low–crank capability must be measured.



SCHEMATIC REVIEW ONLY – PCB/HIL release gates remain open

digitie/canview

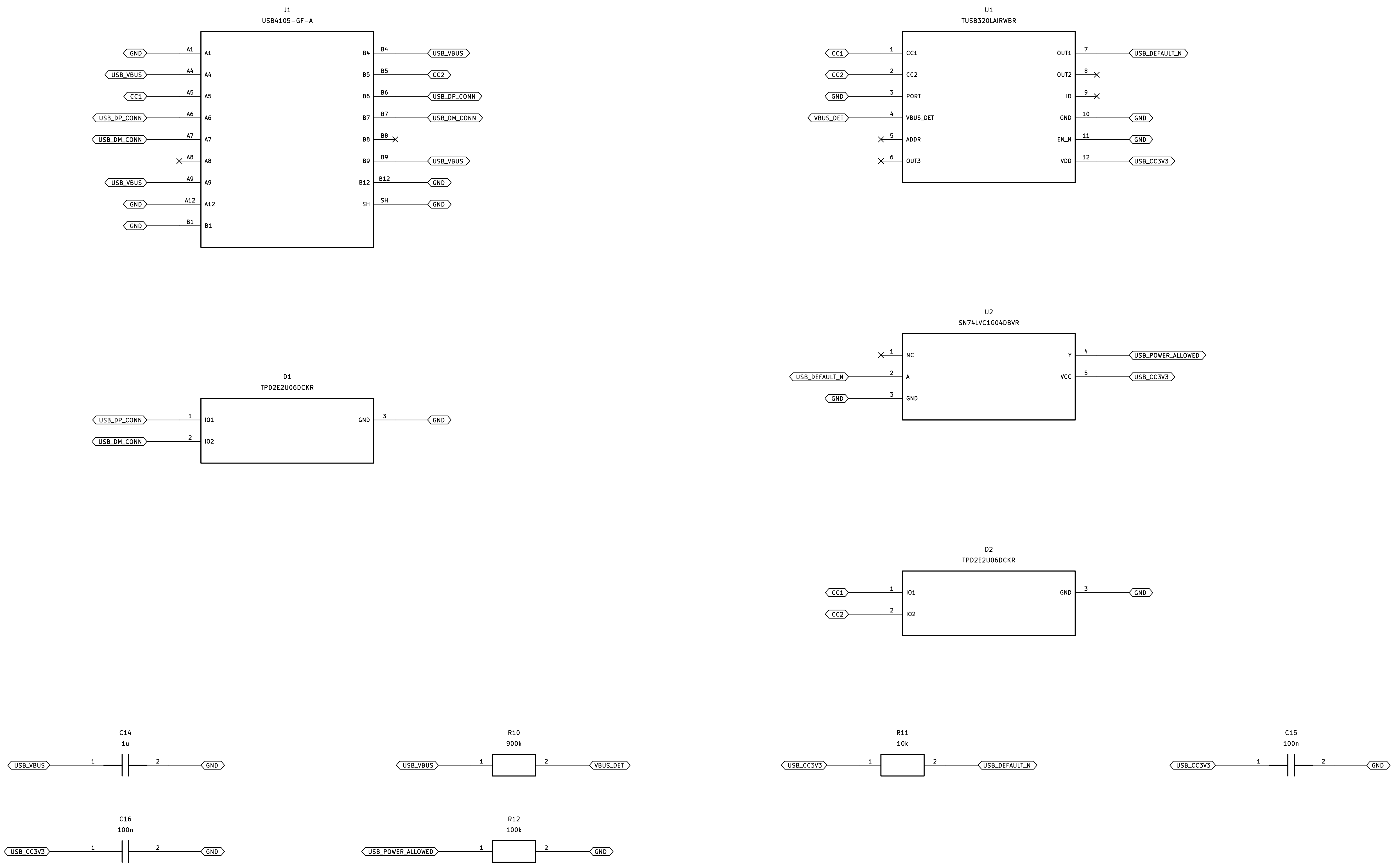
Sheet: /00c_auto_5v/

File: 00c_auto_5v.kicad_sch

Title: CANView / 00c_auto_5v

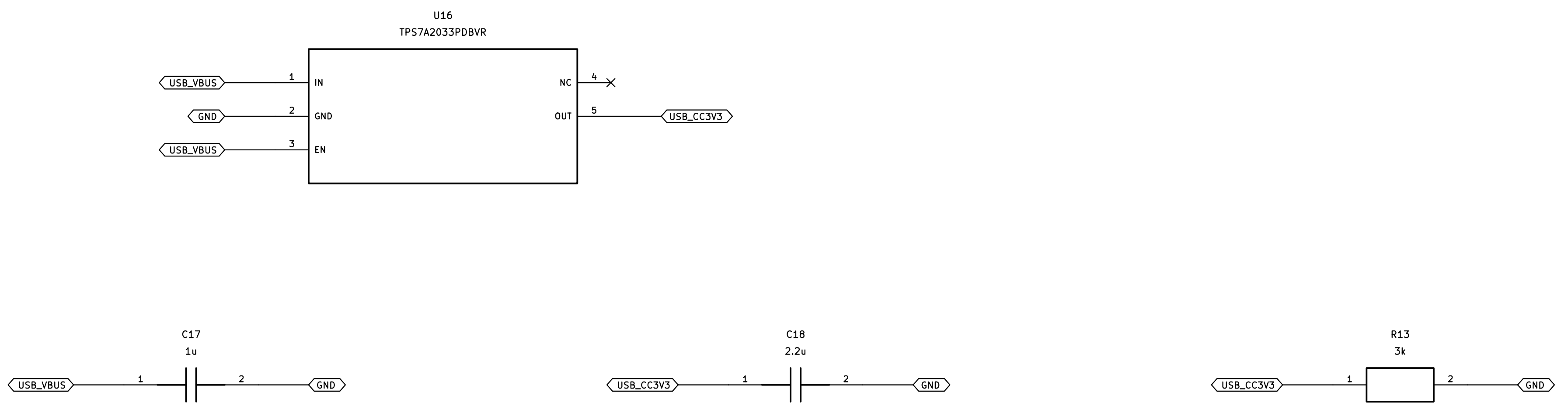
Size: A3	Date: 2026-09-05	**Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 4/29

USB-C sink: requires CC advertisement >=1.5A. Default-current ports DO NOT power the board. No USB PD/high-voltage request.



SCHEMATIC REVIEW ONLY – PCB/HIL release gates remain open
digitie/canview
Sheet: /01_usb_c/
File: 01_usb_c.kicad_sch
Title: CANView / 01_usb_c
Size: A3 | Date: 2026-09-05 | **Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6 | Id: 5/29

Pre−limiter USB−only3.3V powers CC detector and its inverter. TUSB320LAI VDD maximum5.0V: NEVER connect it directly to USB VBUS. No SYS rail dependency.



SCHEMATIC REVIEW ONLY − PCB/HIL release gates remain open
digitie/canview
Sheet: /01a_usb_cc_supply/
File: 01a_usb_cc_supply.kicad_sch

Title: CANView / 01a_usb_cc_supply

Size: A3	Date: 2026-09-05	**Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 6/29

USB load current limit about1A, below the minimum1.5A CC advertisement. Input<=6.5V; no PD sink negotiation.

SCHEMATIC REVIEW ONLY – PCB/HIL release gates remain open

digitie/canview

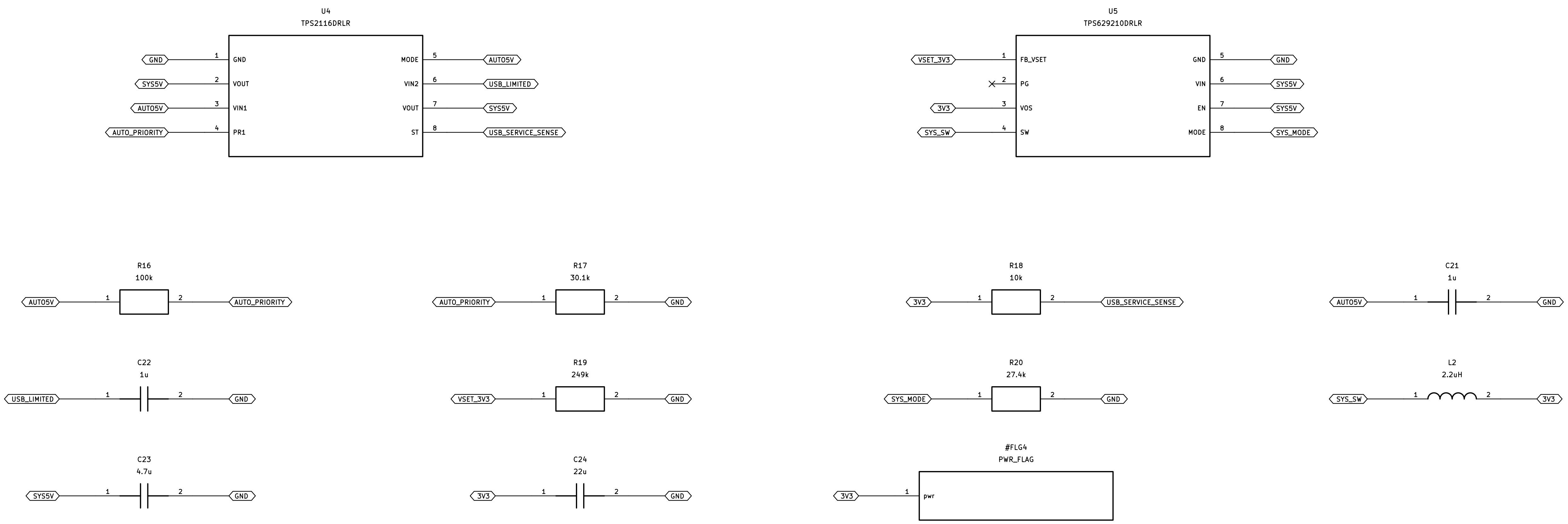
Sheet: /01b_usb_limit/

File: 01b_usb_limit.kicad_sch

Title: CANView / 01b_usb_limit

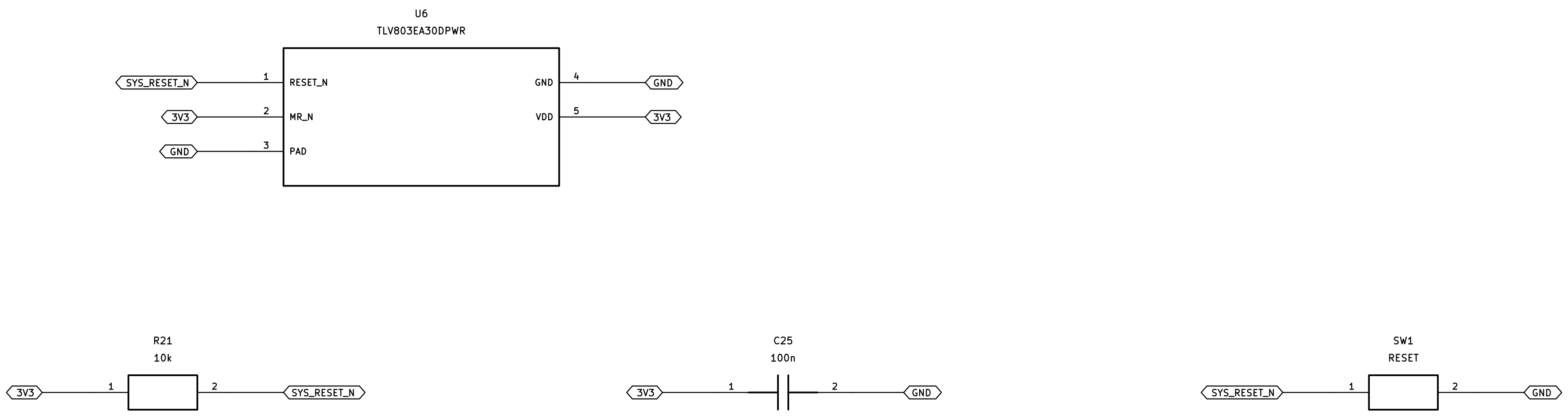
Size: A3	Date: 2026-09-05	**Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 7/29

TPS2116 isolates automotive5V and USB; CAN PHY/GPS never fed from mux output.3.3V EN follows SYS5V, not automotive PGOOD.



SCHEMATIC REVIEW ONLY — PCB/HIL release gates remain open
digitie/canview
Sheet: /02_system_power/
File: 02_system_power.kicad_sch
Title: CANView / 02_system_power
Size: A3 Date: 2026-09-05 **Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6 Id: 8/29

MCU supervisor3.08V with130..270ms release. Communicator also has an independent PHY-domain supervisor.



SCHEMATIC REVIEW ONLY – PCB/HIL release gates remain open

digitie/canview

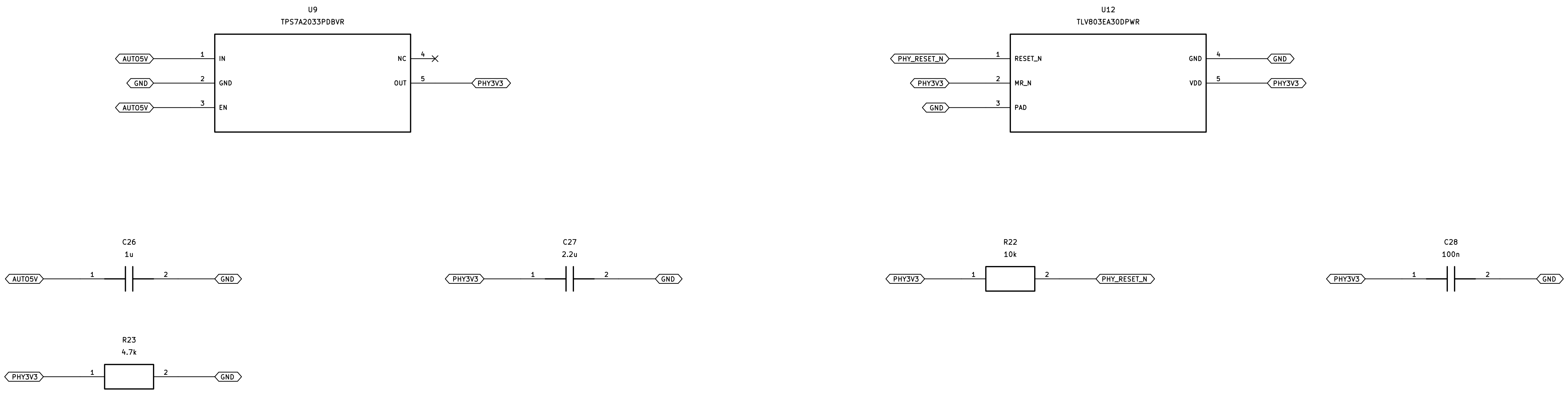
Sheet: /03_reset/

File: 03_reset.kicad_sch

Title: CANView / 03_reset

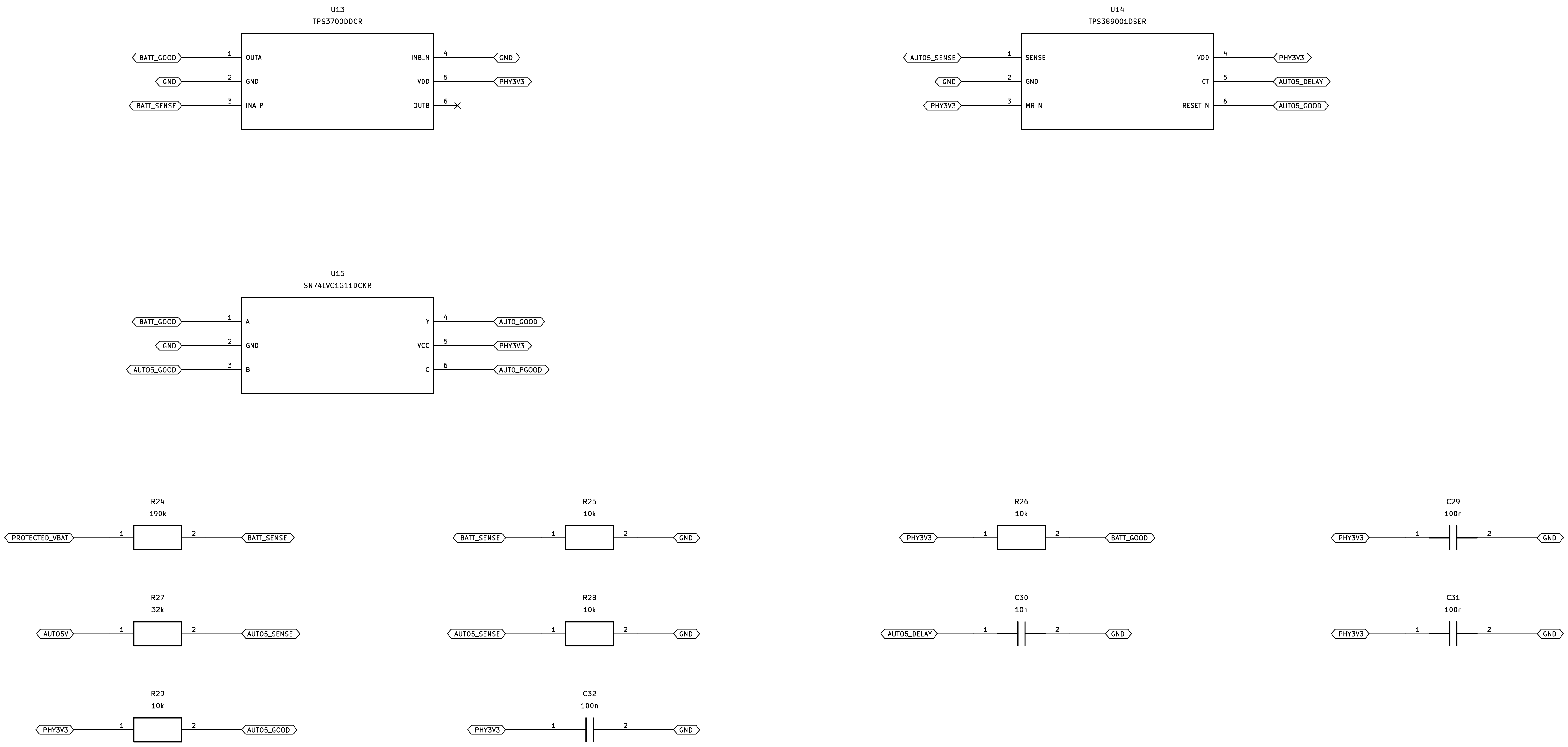
Size: A3	Date: 2026-09-05	**Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 9/29

PHY3V3 is automotive-only. Its own supervisor initializes arm latch on USB-first hot-plug. FT inputs track AUTO5V, not held-up3.3V.



SCHEMATIC REVIEW ONLY — PCB/HIL release gates remain open
digitie/canview
Sheet: /03b_phy_power/
File: 03b_phy_power.kicad_sch
Title: CANView / 03b_phy_power
Size: A3 | Date: 2026-09-05 | **Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6 | Id: 10/29

Independent battery threshold and real5V qualification. PGOOD alone is not sufficient for MAX3055. Threshold/latency worst cases remain test gates.



SCHEMATIC REVIEW ONLY – PCB/HIL release gates remain open

digitie/canview

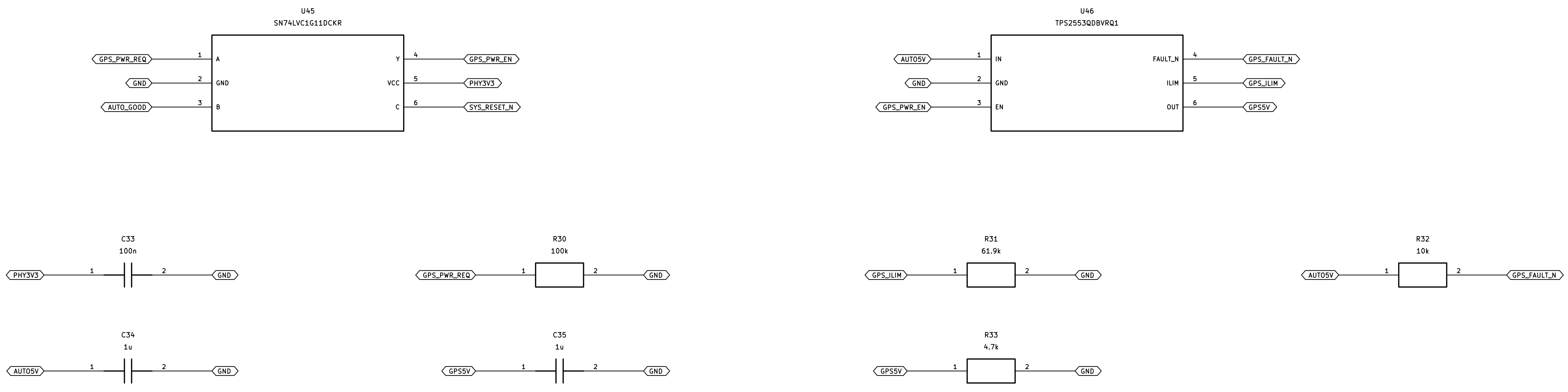
Sheet: /03c_auto_qualification/

File: 03c_auto_qualification.kicad_sch

Title: CANView / 03c_auto_qualification

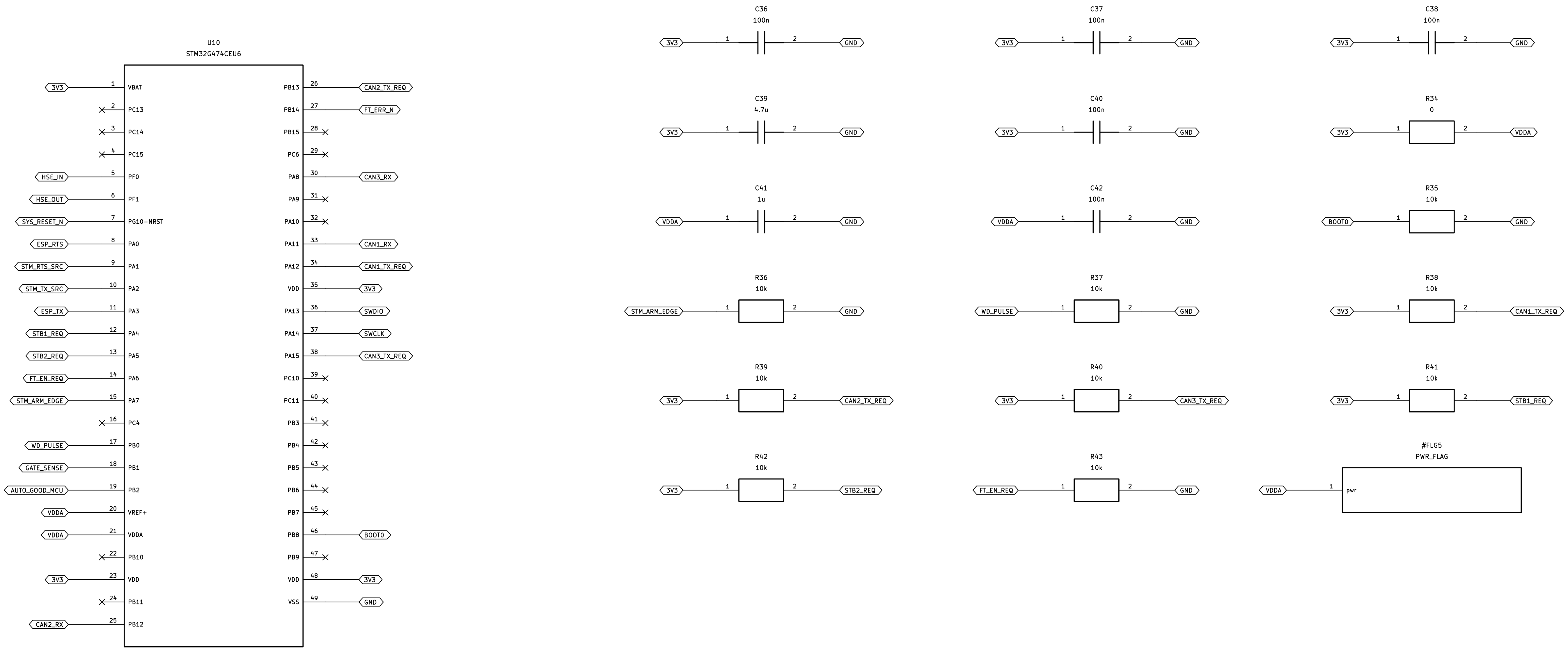
Size: A3	Date: 2026-09-05	**Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 11/29

GPS automotive–only switched supply. Fault–current limit independent of ESP. Never enable in USB–only service.



SCHEMATIC REVIEW ONLY – PCB/HIL release gates remain open
digitie/canview
Sheet: /19_gps_power/
File: 19_gps_power.kicad_sch
Title: CANView / 19_gps_power
Size: A3 | Date: 2026–09–05 | **Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6 | Id: 12/29

STM32G474CEU6 / UFQFPN48. Pin 7 is the PG10-NRST dual-function reset pad. HSE 16MHz. USART2 4Mbps RTS/CTS. SWD only; USB pins are CAN1.



SCHEMATIC REVIEW ONLY – PCB/HIL release gates remain open

digitie/canview

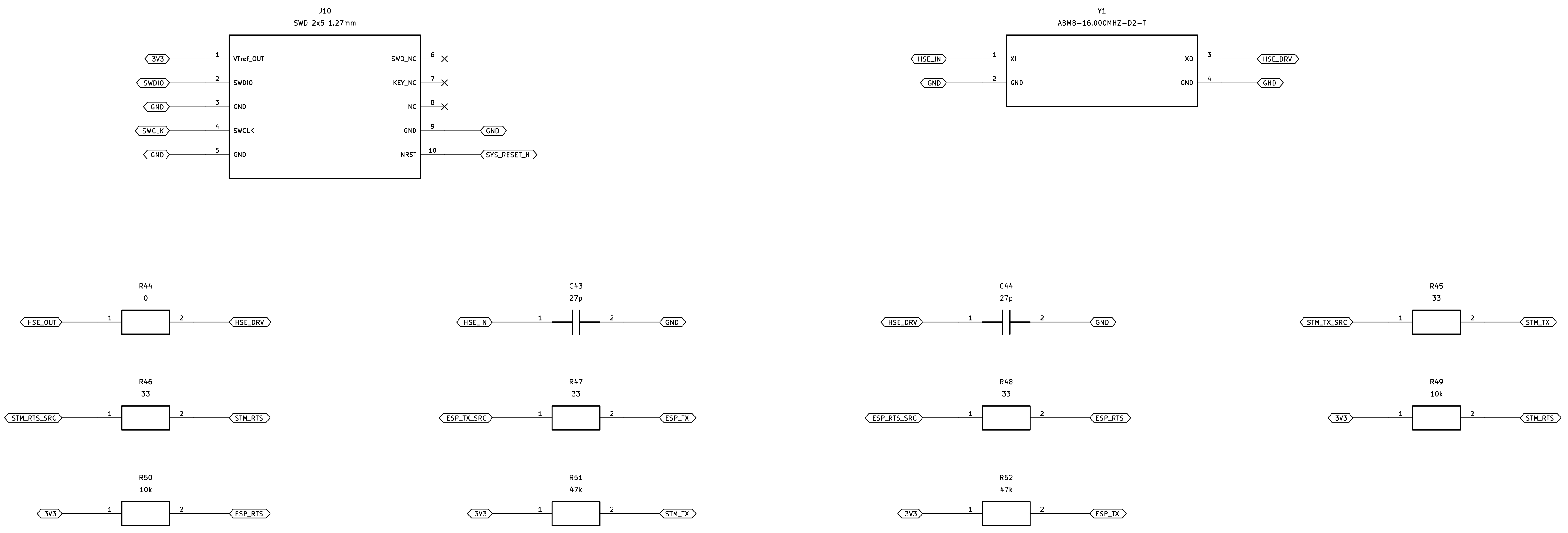
Sheet: /04_stm32/

File: 04_stm32.kicad_sch

Title: CANView / 04_stm32

Size: A3	Date: 2026-09-05	**Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 13/29

16MHz crystal CL18pF: 27pF + estimated 4.5pF parasitic; tune after PCB. SWD VTref is OUTPUT ONLY.

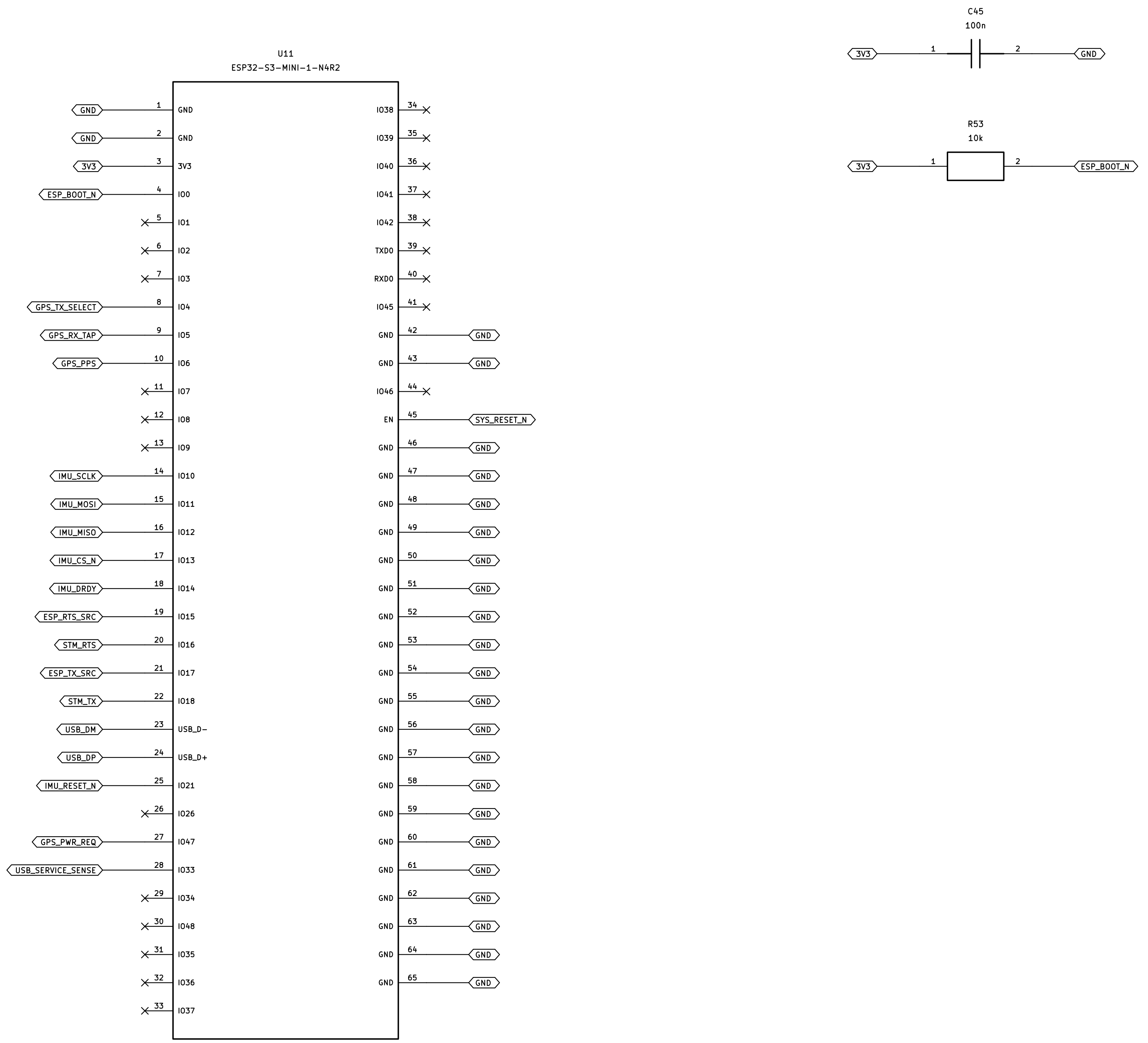


SCHEMATIC REVIEW ONLY – PCB/HIL release gates remain open
digitie/canview
Sheet: /05_debug_clock_uart/
File: 05_debug_clock_uart.kicad_sch

Title: CANView / 05_debug_clock_uart

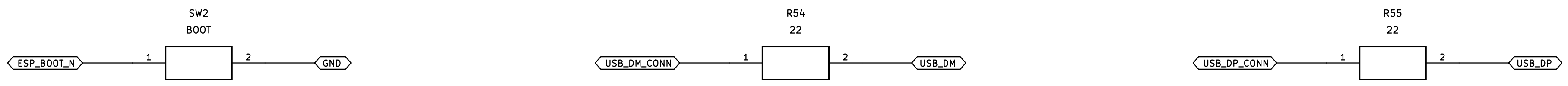
Size: A3	Date: 2026-09-05	**Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 14/29

ESP32-S3-MINI-1. Keep antenna clearance. USB native serial/JTAG. PSRAM pins reserved.



SCHEMATIC REVIEW ONLY – PCB/HIL release gates remain open
digitie/canview
Sheet: /06_esp32/
File: 06_esp32.kicad_sch
Title: CANView / 06_esp32
Size: A3 Date: 2026-09-05 **Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6 Id: 15/29

BOOT button for ROM download. Reset supervisor is shared on Communicator. Keep GPIO0 released for normal boot.



SCHEMATIC REVIEW ONLY — PCB/HIL release gates remain open

digitie/canview

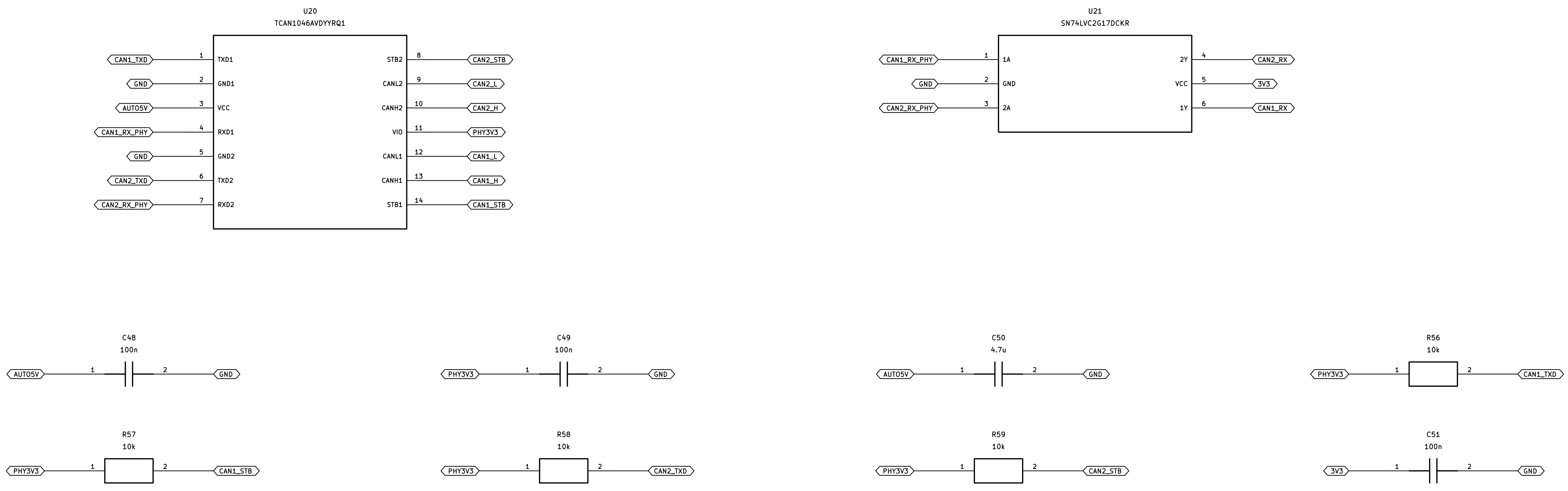
Sheet: /07_esp_service/

File: 07_esp_service.kicad_sch

Title: CANView / 07_esp_service

Size: A3	Date: 2026-09-05	**Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 16/29

TCAN1046AV: actual pin11=VIO, pin5=GND2, pin7=RXD2, pin8=STB2. CAN PHY power is vehicle-only.

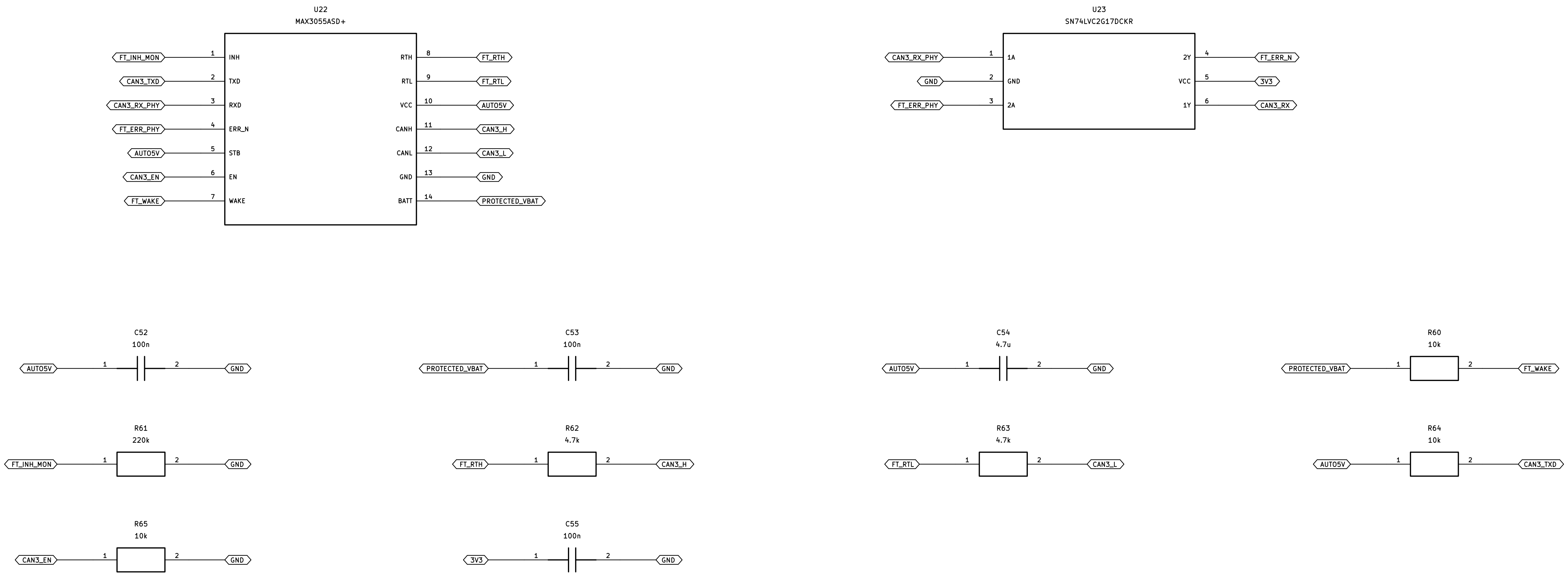


SCHEMATIC REVIEW ONLY – PCB/HIL release gates remain open
digitie/canview
Sheet: /08_can_fd/
File: 08_can_fd.kicad_sch

Title: CANView / 08_can_fd

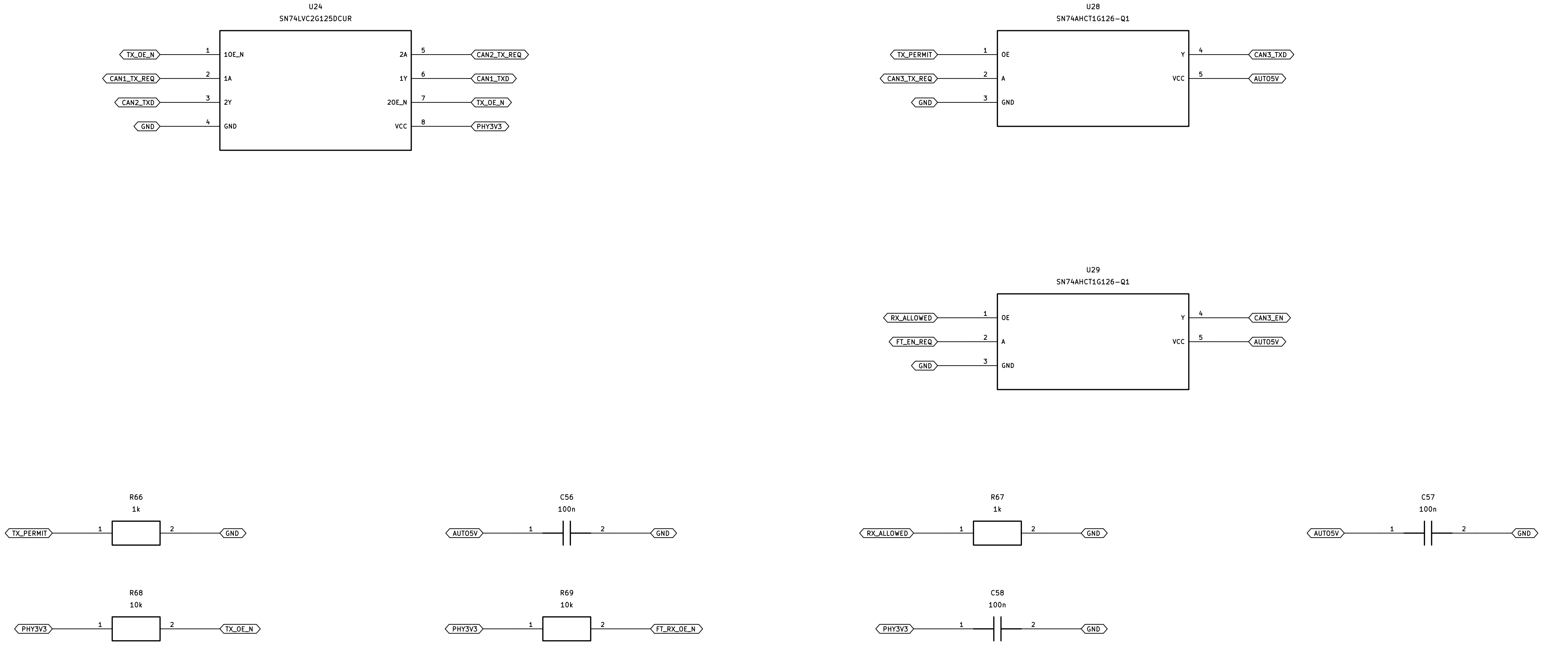
Size: A3	Date: 2026-09-05	**Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 17/29

MAX3055 only for verified ISO11898-3 125kbps bus. RTH/RTL 4.7k distributed termination; no CANH-CANL 120ohm.



SCHEMATIC REVIEW ONLY – PCB/HIL release gates remain open
digitie/canview
Sheet: /09_can_ft/
File: 09_can_ft.kicad_sch
Title: CANView / 09_can_ft
Size: A3 | Date: 2026-09-05 | **Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6 | Id: 18/29

Independent TX gate: no permit => tri-state TXD + PHY rail pull-ups. Ioff input buffers prevent USB-only backpower.



SCHEMATIC REVIEW ONLY – PCB/HIL release gates remain open

digitie/canview

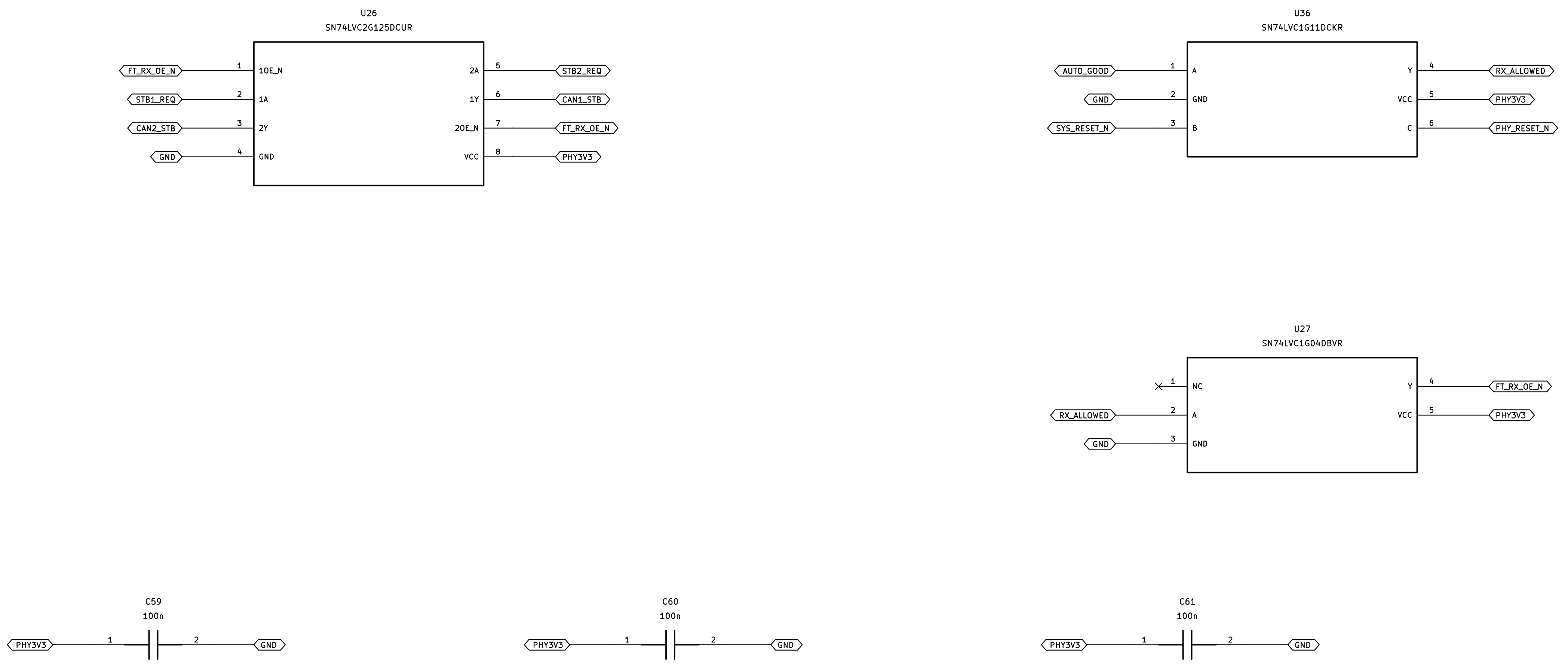
Sheet: /10_tx_gate/

File: 10_tx_gate.kicad_sch

Title: CANView / 10_tx_gate

Size: A3	Date: 2026-09-05	**Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 19/29

RX mode is independent of TX permit. On invalid automotive power all PHY enables return to hardware standby.



SCHEMATIC REVIEW ONLY — PCB/HIL release gates remain open

digitie/canview

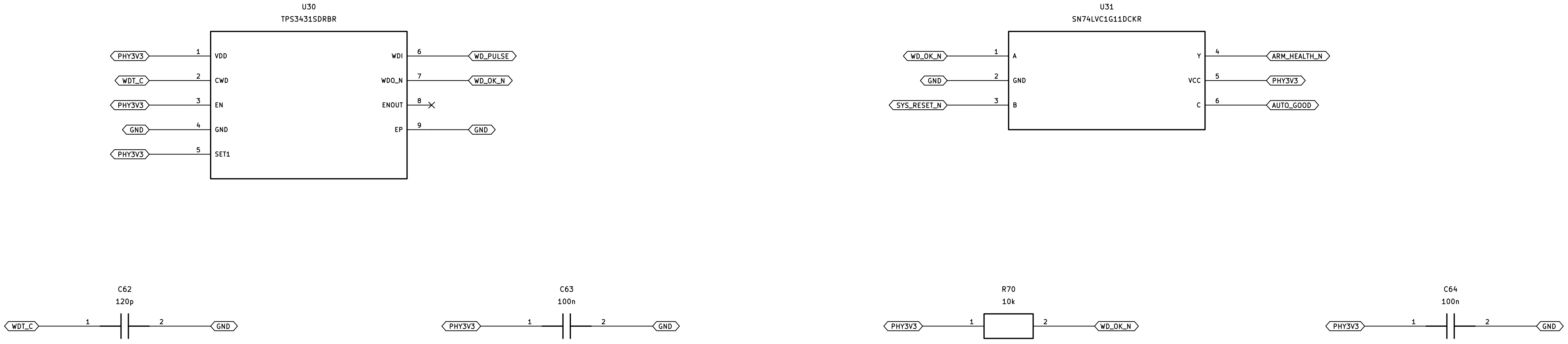
Sheet: /11_phy_modes/

File: 11_phy_modes.kicad_sch

Title: CANView / 11_phy_modes

Size: A3	Date: 2026-09-05	**Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 20/29

120pF gives64.288ms nominal watchdog; budget<72ms with5% C and10pF stray. WDO recovery CANNOT re-arm TX; new ARM edge required.



SCHEMATIC REVIEW ONLY — PCB/HIL release gates remain open

digitie/canview

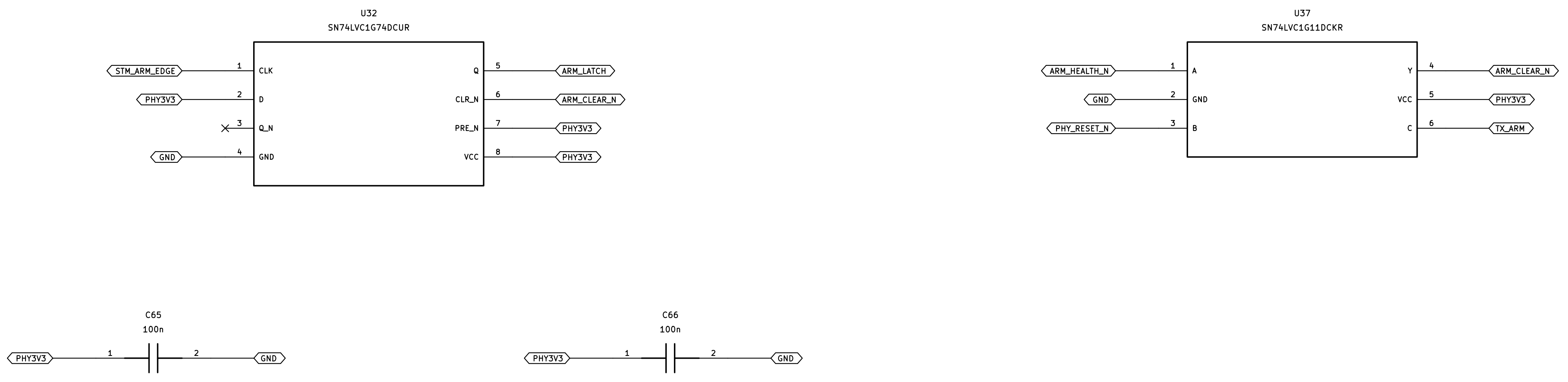
Sheet: /12_watchdog_latch/

File: 12_watchdog_latch.kicad_sch

Title: CANView / 12_watchdog_latch

Size: A3	Date: 2026-09-05	**Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 21/29

PHY–domain POR and physical disarm both CLEAR the latch, including USB–first automotive hot–plug. No stale Q reuse.



SCHEMATIC REVIEW ONLY – PCB/HIL release gates remain open

digitie/canview

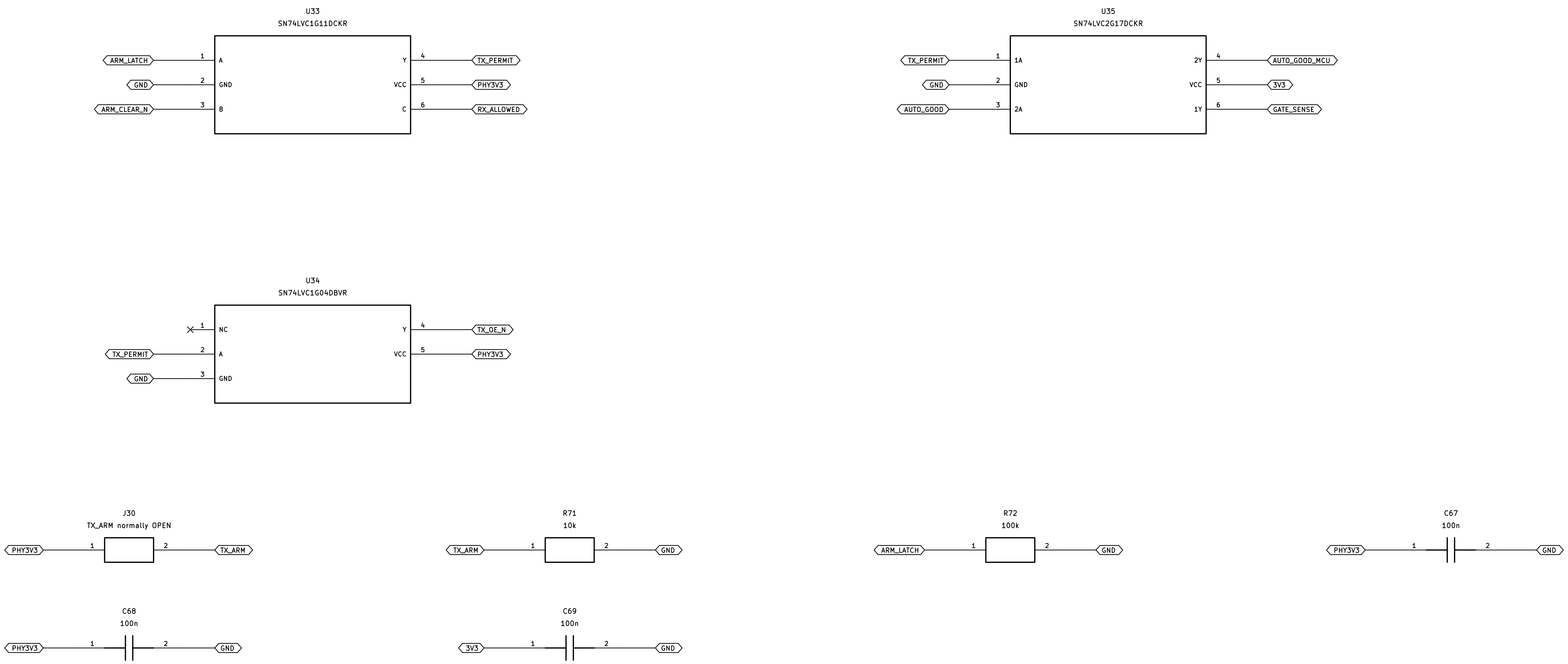
Sheet: /12b_latch_clear/

File: 12b_latch_clear.kicad_sch

Title: CANView / 12b_latch_clear

Size: A3	Date: 2026-09-05	**Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 22/29

Physical arm link is OPEN by default. Software cannot bypass the link. Gate sense is buffered into MCU power domain.



SCHEMATIC REVIEW ONLY — PCB/HIL release gates remain open

digitie/canview

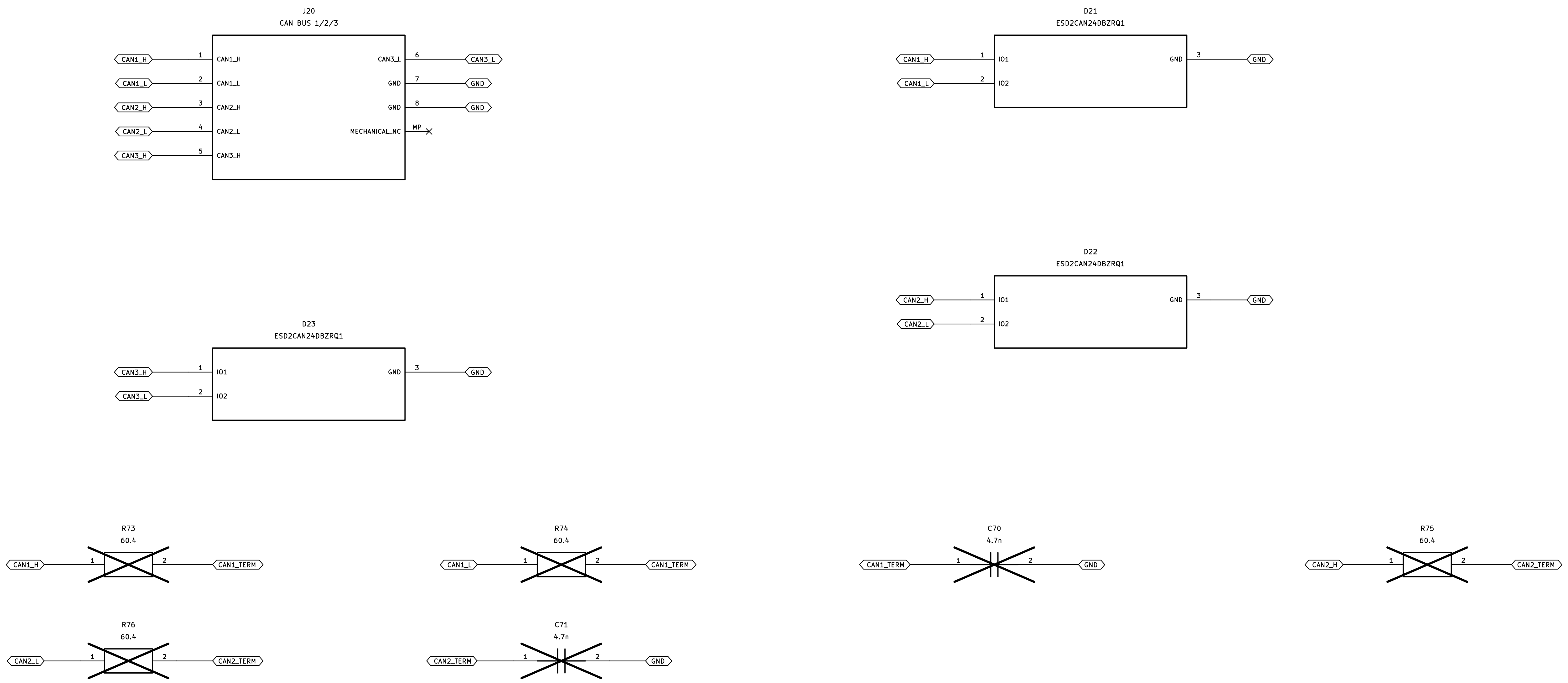
Sheet: /13_arm_switch/

File: 13_arm_switch.kicad_sch

Title: CANView / 13_arm_switch

Size: A3	Date: 2026-09-05	**Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 23/29

BOARD pinout only, NOT a Tucson harness assignment. CAN1/2 split termination DNP for vehicle stub. CAN3 uses RTH/RTL.



SCHEMATIC REVIEW ONLY — PCB/HIL release gates remain open

digitie/canview

Sheet: /14_can_connectors/

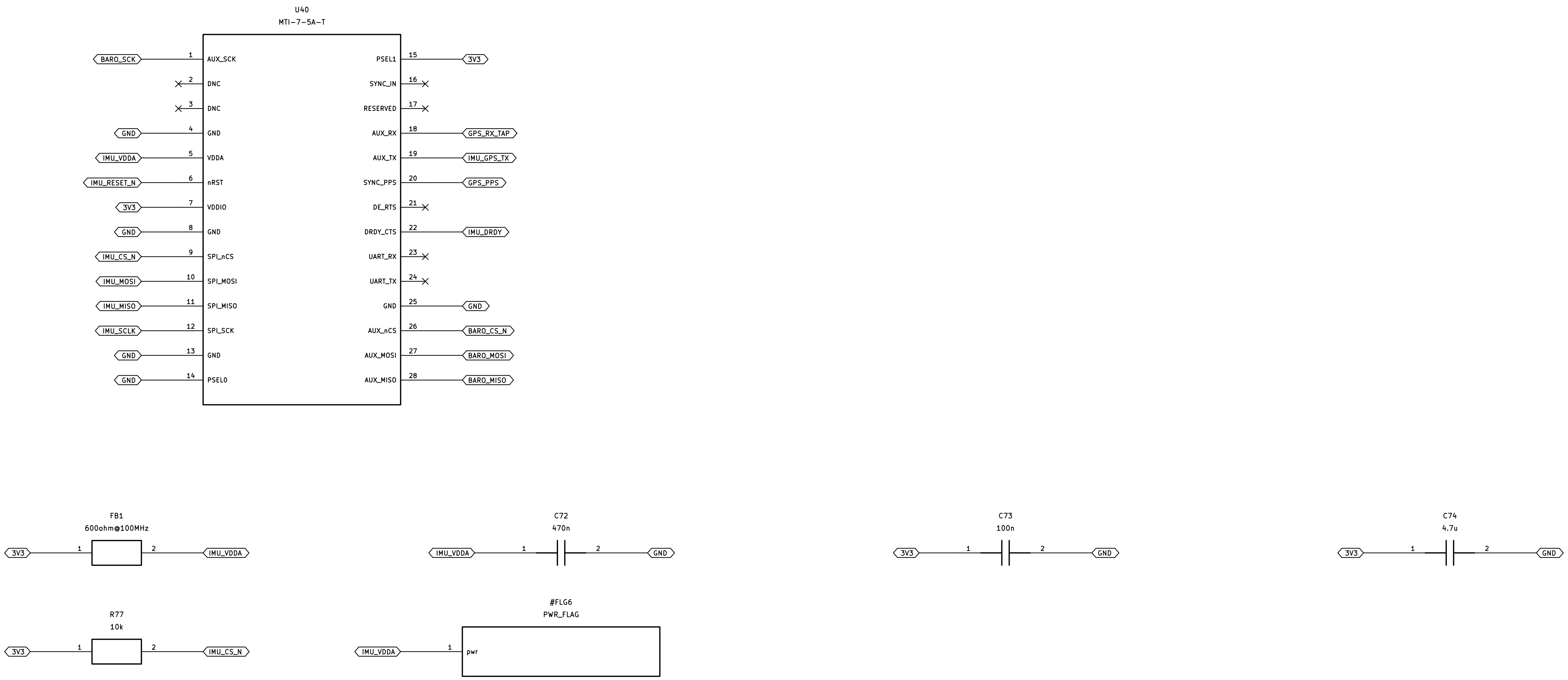
File: 14_can_connectors.kicad_sch

Title: CANView / 14_can_connectors

Size: A3 | Date: 2026-09-05 | **Rev: R1 REVIEW**

KiCad E.D.A. 10.0.6 | Id: 24/29

MTI-7-5A-T v5 module: host SPI mode3 <=2MHz. DNC pads remain NC. GPS/PPS required for qualified GNSS/INS.



SCHEMATIC REVIEW ONLY – PCB/HIL release gates remain open

digitie/canview

Sheet: /15_gnss_ins/

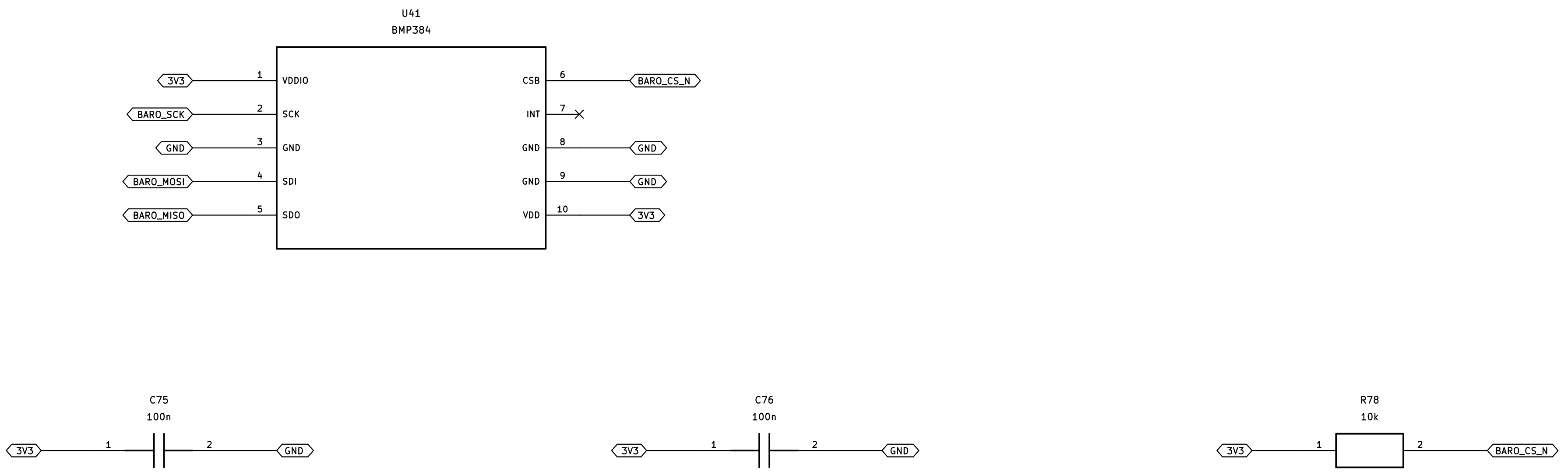
File: 15_gnss_ins.kicad_sch

Title: CANView / 15_gnss_ins

Size: A3 | Date: 2026-09-05 | **Rev: R1 REVIEW**

KiCad E.D.A. 10.0.6 | Id: 25/29

BMP384 barometric aiding belongs to MTi AUX SPI. Require MTi firmware >=1.18. ESP obtains pressure through MTi output.



SCHEMATIC REVIEW ONLY — PCB/HIL release gates remain open

digitie/canview

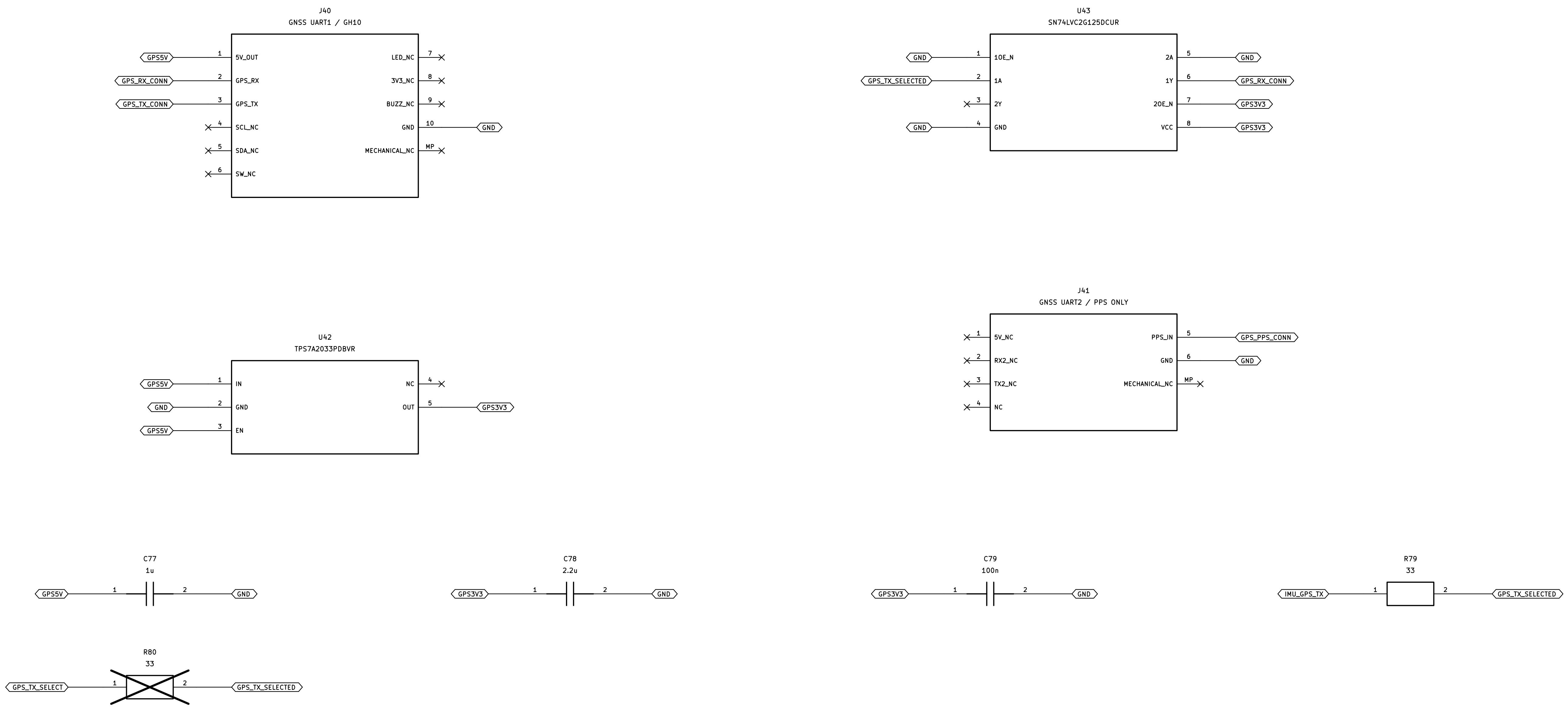
Sheet: /16_barometer/

File: 16_barometer.kicad_sch

Title: CANView / 16_barometer

Size: A3	Date: 2026-09-05	**Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 26/29

Holybro F9P Helical SKU12018: main UART1 GH10 + UART2 PPS GH6. Receiver pin2=RX, pin3=TX. PPS is secondary pin5.



SCHEMATIC REVIEW ONLY – PCB/HIL release gates remain open

digitie/canview

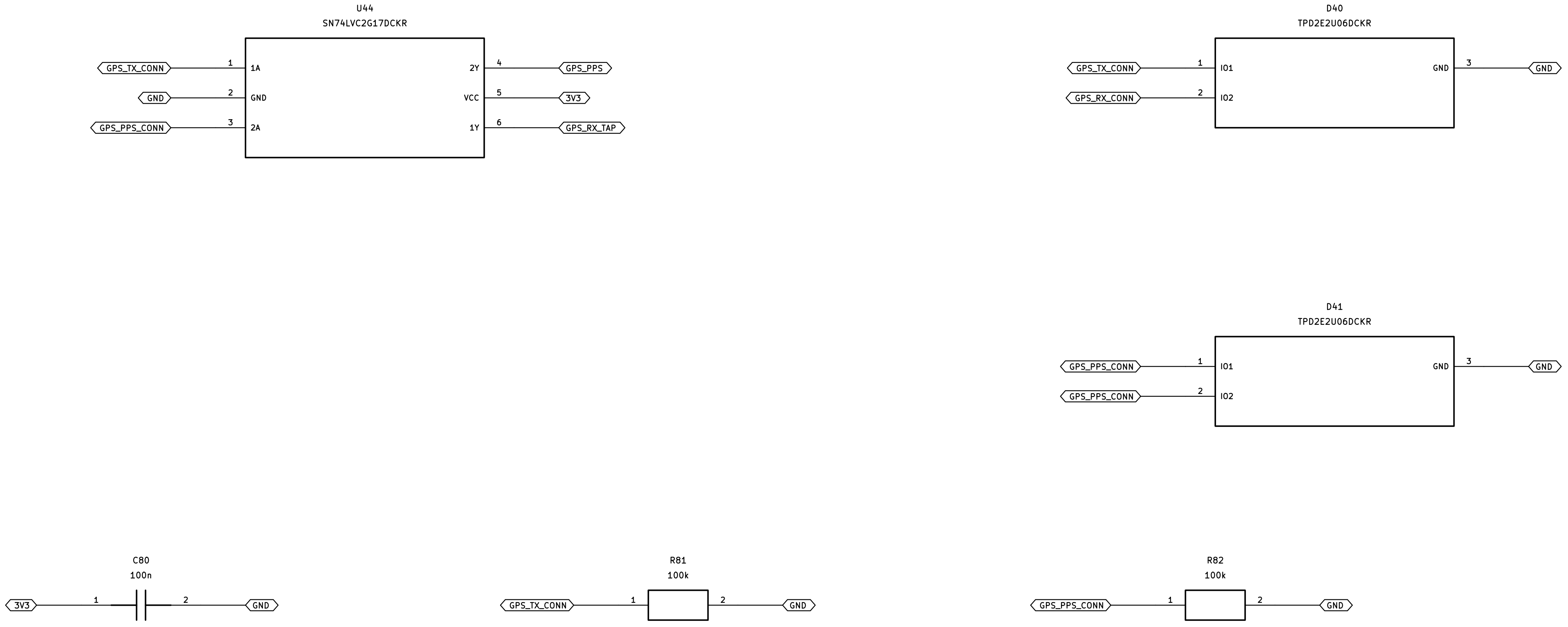
Sheet: /17_gps_ports/

File: 17_gps_ports.kicad_sch

Title: CANView / 17_gps_ports

Size: A3	Date: 2026-09-05	**Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 27/29

Ioff buffers isolate external GPS TX/PPS during local power-off. GPS is disabled in USB service mode.

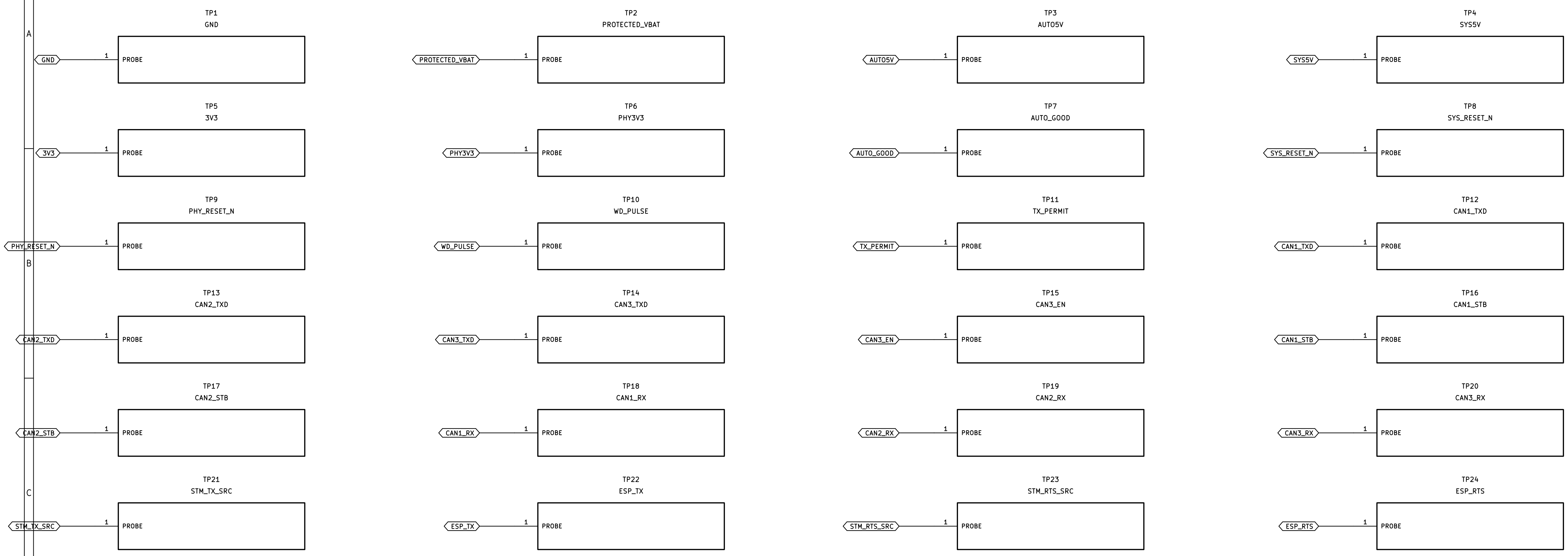


SCHEMATIC REVIEW ONLY — PCB/HIL release gates remain open
digitie/canview
Sheet: /18_gps_protection/
File: 18_gps_protection.kicad_sch

Title: CANView / 18_gps_protection

Size: A3	Date: 2026-09-05	**Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 28/29

1mm probe pads: no fitted connector required. Scope grounds first; never connect earth–referenced probes to an unknown vehicle ground.



SCHEMATIC REVIEW ONLY – PCB/HIL release gates remain open
digitie/canview
Sheet: /20_test_points/
File: 20_test_points.kicad_sch
Title: CANView / 20_test_points

Size: A3	Date: 2026-09-05	**Rev: R1 REVIEW**
KiCad E.D.A. 10.0.6		Id: 29/29